

Research Methodology in CS (CSEG3060_3)

Lecture 5: Plagiarism Prevention Strategies: proper paraphrasing, citation styles, reference management, and Turnitin/URKUND usage

Unit 3: Ethical Conduct and Literature Studies • Mapped Course Outcome: CO3

Lecture 5: Plagiarism Prevention Strategies

Proper Paraphrasing, Citation Styles, Reference Management, and Plagiarism Detection Tools

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** V **Unit:** III — Ethical Conduct and Literature Studies **Lecture Duration:** 60 minutes **Mapped Course Outcome:** CO3 — Demonstrate the ability to conduct ethical research and follow academic integrity standards through proper literature analysis, citation practices, and plagiarism avoidance.

1. Introduction

In the previous lectures of Unit III, we examined the foundations of literature studies, the architecture of a literature review, and the broader ethical considerations that govern research in computer science. We learned that research is a cumulative enterprise — built upon the work of others — and that misrepresenting the source of an idea is among the most serious violations of academic integrity.

Lecture 5 transitions from the *philosophy* of ethical research to its *practice*. While knowing that plagiarism is unethical is necessary, it is not sufficient. Researchers must master the operational skills that prevent inadvertent (and intentional) plagiarism. These skills fall into four interrelated domains:

1. **Paraphrasing** — the art of restating ideas in one's own voice while preserving meaning.
2. **Citation Styles** — the formal language that credits sources unambiguously.
3. **Reference Management** — the systematic organization of bibliographic data.
4. **Plagiarism Detection Tools** — the technological verification layer (Turnitin, URKUND) that confirms originality.

This lecture equips students with the technical competence required to produce original, traceable, and ethically defensible scholarly writing — a competency evaluated directly under **CO3**.

2. Learning Objectives

By the end of this lecture, students should be able to:

#	Objective	Bloom's Level
LO1	Differentiate between acceptable paraphrasing and disguised copying.	Understand
LO2	Apply at least two major citation styles (APA, IEEE) correctly.	Apply
LO3	Use a reference management tool (Mendeley or Zotero) to organize sources and generate bibliographies.	Apply
LO4	Interpret similarity reports from Turnitin/URKUND and act upon them.	Analyze
LO5	Formulate a personal anti-plagiarism workflow for thesis and research paper writing.	Create

3. Key Concepts at a Glance

- **Paraphrase:** Restatement of text in different words and structure while preserving meaning.
 - **Citation:** In-text reference acknowledging the source of an idea, finding, or quotation.
 - **Reference List / Bibliography:** Comprehensive list of all sources cited in a document.
 - **Reference Management Software:** Tools (Mendeley, Zotero, EndNote) that automate citation capture and bibliography generation.
 - **Similarity Index:** Percentage of text in a submission that matches text in comparison databases.
 - **Originality Report:** Annotated output of plagiarism detection software indicating matched sources and overlap statistics.
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4. Detailed Explanations

4.1 The Anatomy of Plagiarism

Before discussing prevention, we must precisely define what we are preventing. Plagiarism, in academic policy (UPES Academic Integrity Code, UGC Guidelines 2018), refers to the appropriation of another person's ideas, processes, results, or words without giving proper credit. It is classified into:

Type	Description	Severity
Direct Plagiarism	Verbatim copying without quotation marks or citation.	Most severe
Self-Plagiarism	Reusing one's own previously published work without disclosure.	Moderate to severe
Mosaic / Patchwork Plagiarism	Mixing phrases from multiple sources without citation.	Severe
Accidental Plagiarism	Unintentional misuse of source material due to poor note-taking.	Moderate
Idea Plagiarism	Using another's conceptual framework without attribution.	Severe

Why it matters in CS: Computer science papers frequently include algorithms, pseudocode, and architectural diagrams. Copying an algorithm description verbatim from a paper and presenting it as one's own contribution constitutes plagiarism, even if the words are modified slightly. The intellectual contribution (the *idea* of the algorithm) belongs to the original author.

4.2 Paraphrasing: Beyond Word Substitution

4.2.1 What Paraphrasing Is — and Is Not

A common misconception among students is that paraphrasing means replacing a few words with synonyms while leaving the sentence structure intact. This practice, often called **patchwriting**, is a form of plagiarism because it preserves the original author's syntax and voice without attribution.

A genuine paraphrase:

- **Changes the sentence structure** completely.

- **Uses vocabulary** appropriate to the writer's own discipline and audience.
- **Maintains fidelity** to the original meaning (without distortion).
- **Includes a citation** to the original source.

4.2.2 The Five-Step Paraphrasing Method

Students should internalize the following procedure, often attributed to academic writing guides (e.g., Purdue OWL, University of Wisconsin Writing Center):

1. **Read and comprehend** the original passage thoroughly.
2. **Set aside** the original text so it cannot be visually copied.
3. **Restate** the idea in your own words and sentence structures.
4. **Compare** your paraphrase with the original to ensure accuracy and originality.
5. **Cite** the source, even though you have not used the original words.

4.2.3 Worked Example

Original Text (from Goodfellow, Bengio, and Courville, 2016, p. 96):

"Deep learning is a particular kind of machine learning that achieves great power and flexibility by learning to represent the world as a nested hierarchy of concepts, with each concept defined in relation to simpler concepts."

Unacceptable Patchwriting (plagiarism risk):

"Deep learning is a specific type of machine learning that achieves high power and flexibility by learning to represent the world as a nested hierarchy of concepts, where each concept is defined in relation to simpler ones."

Critique: Only synonyms ("specific" for "particular," "high" for "great," "where" for "with") are changed; sentence structure is preserved. This would still be flagged by Turnitin.

Acceptable Paraphrase:

"By modeling reality through layers of increasingly abstract representations — where each higher-level construct is built upon the lower-level ones — deep learning derives its representational capacity from hierarchical feature composition, distinguishing it from classical machine learning paradigms (Goodfellow et al., 2016)."

Critique: Sentence structure inverted, vocabulary domain-appropriate ("representational capacity," "hierarchical feature composition"), meaning preserved, citation included.

4.2.4 Quoting vs. Paraphrasing vs. Summarizing

Technique	When to Use	Length	Citation
Quotation	When the original wording is uniquely authoritative or stylistically essential.	Exact words; usually $\leq$ 40 words.	Required + quotation marks.
Paraphrase	When the idea is important but the wording is not.	Approximately same length as original.	Required (no quotation marks).
Summary	When condensing a long passage into a brief statement.	Substantially shorter than original.	Required (no quotation marks).

Rule of thumb for CS theses: Reserve direct quotations for definitions, formal theorems, or standards language. Use paraphrases for algorithm descriptions and prior-work summaries.

4.3 Citation Styles: The Grammar of Attribution

A citation style is a standardized format for documenting sources. Choosing the correct style — and applying it consistently — is a hallmark of academic rigor. In computer science and engineering, the dominant styles are **APA 7th Edition**, **IEEE Editorial Style**, and occasionally **ACM Reference Format**.

4.3.1 APA 7th Edition (American Psychological Association)

Common in social sciences and increasingly accepted in interdisciplinary CS work (e.g., HCI, educational technology).

In-text citation (paraphrase):

Recent work has demonstrated the efficacy of transformer architectures for low-resource NLP tasks (Devlin et al., 2019).

In-text citation (direct quote, with page number):

"The model architecture is a multi-layer bidirectional Transformer encoder" (Devlin et al., 2019, p. 4171).

Reference list entry:

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, 4171–4186. <https://doi.org/10.18653/v1/N19-1423>

4.3.2 IEEE Editorial Style (Institute of Electrical and Electronics Engineers)

The standard style for most IEEE journals, conferences, and the majority of CS publications.

In-text citation:

Recent work has demonstrated the efficacy of transformer architectures for low-resource NLP tasks [1].

Reference list entry:

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

4.3.3 Comparison Table

Element	APA 7th	IEEE
In-text marker	(Author, Year)	[Number]
Reference ordering	Alphabetical by author	Order of appearance
Title case for articles	Sentence case	Lower case (except proper nouns)
DOI format	https://doi.org/...	doi: 10....
Author format	Last, F. M.	F. Last

4.3.4 When to Use Which Style

- **IEEE** — for any submission to IEEE, ACM, Springer LNCS, Elsevier journals, and most B.Tech/M.Tech theses in Indian universities including UPES.
- **APA** — for HCI, cognitive science, education research, and interdisciplinary work.
- **ACM** — when specifically submitting to ACM venues.

Important: Always verify the citation style required by your target venue or guide. Mixing styles is a common error flagged by evaluators.

4.4 Reference Management: From Chaos to Control

Reference management software (RMS) automates the capture, storage, organization, and citation of bibliographic sources. For a B.Tech student managing 30–80 references across a thesis, this is not a luxury — it is a necessity.

4.4.1 The Big Three Tools

Feature	Mendeley	Zotero	EndNote
Developer	Elsevier (now)	Roy Rosenzweig Center (open-source)	Clarivate
Cost	Free (with optional premium)	Free (open-source)	Paid (institutional license)
PDF Storage	Excellent	Excellent (with web plugin)	Good
Word Integration	Yes	Yes	Yes
Browser Capture	Web Importer	Connector	Web capture
Collaboration	Private groups	Group libraries	Shared libraries
Best For	PDF-heavy workflows	Open-source advocates	Large institutional projects

4.4.2 Reference Management Workflow

A robust anti-plagiarism workflow integrates the four pillars discussed in this lecture. The following six-step procedure is recommended:

1. **Capture** — Install the browser connector (Mendeley Web Importer or Zotero Connector) and use it to save references directly from Google Scholar, IEEE Xplore, ACM DL, SpringerLink, and arXiv.
2. **Organize** — Create folders by chapter or topic. Rename PDFs using a clear convention: `Author_Year_Keyword_Title.pdf`.
3. **Annotate** — Highlight key passages and add notes directly in the PDF within the RMS. This builds the conceptual scaffolding for paraphrasing.
4. **Cite While Writing** — Insert citations from the RMS Word/LaTeX plugin. Never type citations manually.
5. **Verify** — Cross-check the auto-generated bibliography against the official style guide. RMS tools occasionally miss publication type nuances (e.g., conference vs. journal distinction).
6. **Backup** — Synchronize the library to the cloud weekly. A lost library is a research catastrophe.

4.4.3 LaTeX-Specific Considerations

For students writing theses in LaTeX (common at UPES for CS), reference management tools integrate through **BibTeX** or **biblatex**. Mendeley and Zotero can export **.bib** files, which are then processed by the compiler. Key tools:

- **BibTeX** — legacy but widely supported.
- **biblatex** — modern, more flexible.
- **Better BibTeX (Zotero plugin)** — generates citation keys automatically (`vaswani2017attention`).

A typical BibTeX entry for the BERT paper:

```
@inproceedings{devlin2019bert,
  author    = {Jacob Devlin and Ming-Wei Chang and Kenton Lee and Kristina Toutanova},
  title     = {{BERT}: Pre-training of Deep Bidirectional Transformers for Language Under
  booktitle = {Proceedings of NAACL-HLT},
  year      = {2019},
  pages     = {4171--4186},
  doi       = {10.18653/v1/N19-1423}
}
```

4.5 Plagiarism Detection Tools: Turnitin and URKUND

4.5.1 Purpose and Limitations

Plagiarism detection software does *not* detect plagiarism per se — it detects **textual similarity**. The interpretation of similarity reports is a human responsibility. These tools compare submitted text against three databases:

1. **Internet archive** (billions of web pages, continuously updated).
2. **Published content** (journal articles, conference proceedings, e-books).
3. **Institutional repository** (prior student submissions at the same university).

A match is not plagiarism; an unattributed match *may* be.

4.5.2 Turnitin

Turnitin (turnitin.com) is the most widely deployed commercial similarity-detection platform, integrated into UPES's learning management ecosystem (via Moodle/Canvas).

How it works:

- Submissions are broken into overlapping text segments.
- Each segment is matched against the comparison databases.
- A similarity index is computed: percentage of matched text to total submission length.
- Color-coded highlights link each match to its suspected source.

Interpreting the index:

Similarity Index	Typical Interpretation	Recommended Action
0–10%	Excellent originality.	Acceptable for most work.
11–24%	Acceptable with proper attribution.	Review highlighted sources.
25–50%	Substantial similarity detected.	Significant revision required.
> 50%	Likely unattributed copying.	Rewriting and re-submission mandatory.

Important caveat: A 0% similarity score does *not* guarantee originality. Translated, paraphrased, or non-textual content (code, diagrams) may evade detection. Furthermore, common phrases, boilerplate text, and methodological descriptions naturally produce matches.

UPES Policy Context: Most Indian universities, including UPES, permit a maximum similarity index of **10–20%** for theses (excluding references, quotations, and common methodological phrases). Exceeding this threshold triggers mandatory revision and supervisor review.

4.5.3 URKUND (Now Ouriginal)

URKUND, rebranded as **Ouriginal** (ouriginal.com) after a 2020 merger with PlagScan, is widely used in European and Scandinavian universities and increasingly in Indian institutions.

Distinctive features compared to Turnitin:

- **Cross-language detection** — can identify translated plagiarism.
- **Modular API** — easily integrated into institutional LMS.
- **Character-by-character analysis** — finer granularity than Turnitin's n-gram approach.
- **Generative AI detection module** — Ouriginal added detection capabilities for GPT-style outputs in 2023.

Typical workflow:

Submission → LMS upload → URKUND server → Document parsed →

Compared against:

- Internet archive
- Published databases (CrossRef, etc.)
- Institutional repository
 - Similarity report generated
 - Delivered to instructor in < 30 minutes

4.5.4 Comparison of Major Tools

Feature	Turnitin	URKUND/Ouriginal	iThenticate	PlagiarismChecker
Market	Academic (student work)	Academic + Research	Research/Publication	General
Color-coded report	Yes	Yes	Yes	Basic
Excludes references	Yes (configurable)	Yes	Manual	No
AI content detection	Yes (since 2023)	Yes (Ouriginal)	Limited	No
Cost	Institutional license	Institutional license	Per-check	Free tier
UPES availability	Yes	Limited	Limited	Yes (self-check)

4.5.5 Acting on the Similarity Report

Receiving a similarity report is not the end of the process — it is the beginning. The following protocol is recommended:

1. **Review each highlighted segment** individually.
2. **Identify the source** by clicking on the match.
3. **Classify each match** as one of:
 - Properly cited quotation (acceptable).
 - Properly paraphrased with citation (acceptable).
 - Common knowledge or methodological phrase (acceptable; exclude).

- Reference list entries (exclude).
- **Unattributed textual overlap (action required).**

4. **Revise** the offending passages using the paraphrasing techniques discussed in Section 4.2.
 5. **Re-submit** and verify that the similarity index has dropped below the institutional threshold.
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5. Practical Examples and Case Studies

Case Study 5.1: The Patchwriting Trap

Scenario: A student is writing the Related Work section of a thesis on federated learning. They read McMahan et al. (2017) and draft their section by replacing some words with synonyms.

Original (McMahan et al., 2017):

"Federated learning is a machine learning setting where many clients collaboratively train a model under the coordination of a central server, while keeping their training data decentralized."

Student's draft:

"Federated learning is a setting of machine learning where many clients train a model collaboratively under the coordination of a central server, while keeping training data decentralized."

Diagnosis: Sentence structure preserved; only minor synonym substitution ("a machine learning setting" → "a setting of machine learning"). This is patchwriting and will be flagged.

Corrected paraphrase with citation:

The federated learning paradigm distributes model training across numerous client devices, each retaining local data, while a central coordinator aggregates updates without direct access to the underlying datasets (McMahan et al., 2017).

Case Study 5.2: Citation Style Drift

Scenario: A student begins writing in APA style for a coursework assignment and switches to IEEE mid-document for the project thesis. The bibliography contains mixed entries.

Consequence: IEEE reviewers will immediately reject the submission, and even APA-only evaluators will flag inconsistency.

Resolution: Establish the citation style at the start of the document and configure the reference management software to use that style exclusively. Verify the bibliography against the official style guide before submission.

Case Study 5.3: Interpreting a Turnitin Report

Scenario: A student's thesis chapter returns a similarity index of 27%. The instructor suspects a problem.

Investigation reveals:

- 8% from properly cited quotations.
- 12% from methodological phrases common to all CS theses (e.g., "experimental setup was designed to evaluate...").
- 5% from reference list titles (excludable).
- 2% from one uncited paragraph lifted from a survey paper.

Corrected index (excluding quotations, common phrases, references): 2%. The student must rewrite the uncited paragraph with proper attribution, but the thesis is otherwise in good standing.

Pedagogical lesson: Always compute the *effective* similarity index, excluding acceptable categories. Tools provide filters for this purpose.

6. Summary

This lecture operationalized the principles of ethical research by examining four interconnected competencies:

1. **Paraphrasing** is a cognitive skill requiring comprehension, restructuring, and citation — not mere synonym substitution. The five-step method (read, set aside, restate, compare, cite) is the foundation.
2. **Citation styles** are formal languages that vary by discipline. IEEE dominates CS; APA is increasingly used in interdisciplinary work. Style consistency is non-negotiable.

3. **Reference management software** (Mendeley, Zotero, EndNote) eliminates manual citation errors and enables seamless integration with Word and LaTeX. A systematic six-step workflow — capture, organize, annotate, cite, verify, backup — is recommended.
4. **Plagiarism detection tools** (Turnitin, URKUND/Ouriginal) produce similarity reports that require human interpretation. An index below 10–20%, after excluding references, quotations, and methodological phrases, indicates compliance with UPES standards.

Together, these competencies form the **operational layer of academic integrity** and are central to achieving **CO3** of this course.

7. Exam Preparation Tips and Practice Questions

7.1 Preparation Tips

- **Memorize** the structure of one citation style in detail (preferably IEEE for CS). Be able to produce both an in-text citation and a reference list entry from memory.
- **Practice** paraphrasing three short technical passages using the five-step method.
- **Install** Mendeley or Zotero before the next assignment; build a small library of 5–10 references.
- **Submit** a sample paragraph through a free plagiarism checker (e.g., SmallSEOTools, DupliChecker) before formal submission to identify accidental overlap.
- **Review** the UPES Academic Integrity Code and the specific similarity threshold for theses in your department.

7.2 Short-Answer Questions (2–3 marks each)

1. Differentiate between paraphrasing and patchwriting. Why is the latter ethically problematic?
2. List the four components of a complete IEEE in-text citation and reference entry.
3. What are the three databases against which Turnitin compares submissions?
4. State the UPES-recommended maximum similarity index for a B.Tech thesis, excluding references and quotations.

7.3 Medium-Answer Questions (5 marks each)

5. Explain the five-step paraphrasing method with an example drawn from a CS research paper of your choice.
6. Compare Mendeley and Zotero as reference management tools. Under what circumstances would you prefer one over the other?

7. A student receives a Turnitin similarity index of 32%. Describe, step by step, how they should analyze and act upon this report.

7.4 Long-Answer / Application Questions (10 marks each)

8. **Application:** You are writing the Literature Review chapter of your B.Tech thesis on "Privacy-Preserving Machine Learning in Healthcare." Draft (a) two properly paraphrased passages from a selected paper with IEEE citations, (b) the corresponding IEEE reference entries, and (c) a short justification of your paraphrasing choices.
9. **Analytical:** Critically evaluate the statement: "A high similarity index from Turnitin always indicates plagiarism." Discuss the limitations of similarity-detection tools and the importance of human interpretation. Cite at least two scholarly references in your answer.
10. **Constructive:** Design a personal anti-plagiarism workflow for a B.Tech thesis project, integrating paraphrasing technique, citation style, reference management, and similarity checking. Justify each step with reference to ethical principles discussed in Unit III.
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8. Suggested Reading

1. American Psychological Association. (2020). *Publication Manual of the American Psychological Association* (7th ed.). APA.
 2. IEEE Editorial Style Manual. (2022). IEEE. <https://www.ieee.org/content/dam/ieee-org/ieee/web/org/pubs/ieee-style-manual.pdf>
 3. Roig, M. (2015). *Avoiding plagiarism, self-plagiarism, and other questionable writing practices: A guide to ethical writing*. Office of Research Integrity. <https://ori.hhs.gov/avoiding-plagiarism>
 4. University of Wisconsin-Madison Writing Center. (2023). *Paraphrasing*. <https://writing.wisc.edu/handbook/assignments/reusing-paraphrasing/>
 5. UPES. (2023). *Academic Integrity and Plagiarism Policy*. UPES Student Handbook.
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End of Lecture 5 Notes — Unit III, Research Methodology in Computer Science (CSEG3060_3).